## INTRODUCTION

A building is a living place surrounded by walls and covered by roof for the purpose of keeping out rain, sun, wind and snow. It may be a bungalow, apartment, school, hospital, shopping complex, industry, residential building etc. Any building essentially comprises of three parts namely foundation, super structure and roof. Before construction the civil engineer has to plan and prepare the building drawing with all details.

The main aim of building drawing is to give sufficient informations by the designer to the construction engineer. In order to give sufficient information about the building the following views are generally drawn:

**A.** Plan

**B.** Elevation and

**C.** Section

### A. PLAN

The building is imagined to be cut by a horizontal plane at the sill level of the window. The upper portion is removed. Now building is seen from top. A projection of the remaining portion of the building on a horizontal plane is known as the plan.

### B. ELEVATION

It is the front view of building. Imagine to stand in front of it. Whatever the portion of the building is visible above the ground level, take its first angle projection on a vertical plane behind the building known as the elevation.

### C. SECTION

The building is imagined to be cut by a vertical plane in order to show the internal details such as details of foundation, flooring, doors, windows, ventilators, thickness of walls, lintels, roof, parapet wall, sunshade, etc. Arrows at the extreme ends of the section plane or planes show the directions in which these details are required. The details drawn and marked on a vertical plane, after removing the part of the building behind the cut section is known as the section.

## TERMINOLOGY

### BUILDING MATERIALS

1. **MASONRY:** According to the type of material used for construction, it is called as stone masonry, brick masonry or concrete masonry.
2. **CEMENT MORTAR (C.M):** It is a substance produced from prescribed proportions of cement, sand and water which generally sets hard after mixing and binds the building materials together.
3. **CEMENT CONCRETE (C.C):** It is mixture of cement, sand (Fine aggregate), jelly (coarse aggregate) and water. Concrete mix of 1:1½:3 means 1 part by weight of cement, 1½ parts of sand and 3 parts of jelly are used to form the Mix.
4. **PLAIN CEMENT CONCRETE (PCC):** A plain cement concrete is the concrete without any reinforcement. It is usually referred as cement concrete.
5. **REINFORCED CEMENT CONCRETE (RCC):** It is the concrete reinforced by mild steel or twisted bars.

## BUILDING COMPONENTS:

1. **FOUNDATION:** It is the portion of a building below the ground level (G.L). It transmits the load coming from the superstructure to the ground.
2. **FOOTINGS:** Footings are stepped courses in foundation. These are



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